Response to Reviewers

Round Revision

Manuscript: "Content-Conditioned Invariant Feature-Statistic Augmentation: Unifying Robustness and Weight Averaging for Domain Generalization" (CIFA)
Journal: *Electronics* (MDPI) **Manuscript ID:** electronics-4654208
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Dear Editors and Reviewers,

We thank all three reviewers for the careful and constructive second-round assessment. The remaining concerns are well focused: the readability of the figures (Reviewer 1), the interpretation and positioning of the work together with an editorial cleanup (Reviewer 2), and the mathematical justification, terminology, and internal consistency of the manuscript (Reviewer 3). We have addressed every point; no new training experiments were required.

All revisions are marked in the tracked manuscript (`CIFA_Electronics_tracked.pdf`): additions in blue and deletions in red with a strikethrough. The manuscript uses continuous line numbering (right margin). Each response below is preceded by a location tag of the form [**Tracked PDF: Page X, Lines Y–Z**] that points to the corresponding revision in the tracked PDF; a tag spanning two pages is written as [**Tracked PDF: Page X, Line Y through Page X+1, Line Z**].

The revision comprises five substantive actions:

- **A1.** All seven figures redrawn as publication-ready vector graphics—legends moved outside the plotting area, distinguishable line styles/markers, enlarged fonts and insets, with no change to any data;
- **A2.** Proposition 1 restructured so that the unconditional distribution-level total-variation bounds are separated from the pointwise posterior bound, the latter now stated under an explicit inter-model likelihood-ratio condition (TV $\leq \tanh \alpha$);
- **A3.** The wording made consistent across the paper: non-significant differences are reported as statistically tied/on par, and "equivalence" is used only for the two-one-sided (TOST) test with the equivalence margin stated; no superiority is claimed;
- **A4.** Editorial cleanup, including removal of a duplicated data-availability statement;
- **A5.** A full consistency check of equations, tables, cross-references, and terminology.

We reproduce each comment and respond point by point.

Reviewer 1

Comment. *The remaining concern is the readability of the figures. In several plots, portions of the curves are obscured by legends or overlapping graphical elements, making it difficult to distinguish individual traces and compare the results. The authors should carefully check all figures, reposition legends outside the plotting areas where necessary, and use clearly distinguishable line styles, colors, and markers. Enlarged panels or zoomed-in insets would also help where curves overlap closely. Please ensure that all curves, labels, and legends remain clearly readable at the final publication size. I recommend acceptance after minor revision to address these presentation issues.*

[**Tracked PDF: Figures 1–7, Pages 7, 14, 25, 29, 30, 33, 34**]

Response. We agree, and we have redrawn **all seven figures** (not only the overlapping panels)

as publication-ready vector PDFs, while leaving every data point unchanged. Each of the five requested actions has been applied:

(1) **All figures checked and redrawn.** Figures 1–7 were rebuilt from the identical source data with a uniform print specification;
(2) **Legends moved outside the plotting area.** In-axis legends were relocated to external margins (clearly visible in Figures 2, 6, and 7), so no curve is occluded;
(3) **Distinguishable styles and markers.** Marker size, line width, and cap size were increased and styles differentiated for clear separation at print size;
(4) **Enlarged panels and zoomed insets.** The closely overlapping coverage panel retains a zoomed inset (Figure 2b, zoom 3×10);
(5) **Readability at publication size.** Each figure was re-laid-out at its single- or double-column print dimensions and verified through a print-size preview.

The figure locations are: Figure 1 (Page 7), Figure 2 (Page 14), Figure 3 (Page 25), Figure 4 (Page 29), Figure 5 (Page 30), Figure 6 (Page 33), and Figure 7 (Page 34). We confirm that the underlying values are identical to the authoritative results and that no data were re-sampled.

Reviewer 2

General opening. We thank Reviewer 2 for the candid and careful assessment, and for acknowledging that the presentation issues have been addressed, that the claims have been appropriately revised, and that the null and non-replicated results are reported transparently. The remaining concerns concern the interpretation and positioning of the work rather than additional experiments; we respond to each below.

(1) On the necessity of the distinctive components

[Tracked PDF: Page 33, Lines 1070–1074; Proposition 1, Pages 8–9, Lines 304–348]

We agree that the content–style decomposition is not a source of measured accuracy: full-moment mixing is statistically indistinguishable from full CIFA (paired $+0.23 \pm 0.75$, $p = 0.53$), and removing weight averaging is the dominant ablation (2.42 points). We would, however, distinguish accuracy contribution from theoretical role. The decomposition is not intended to improve accuracy; it is the structural condition that makes the content-preservation bound admissible. By mixing only the style deviation and fixing the content baseline, the construction guarantees $\mathbb{E}[\mathbf{d} \mid Y] = 0$ and thereby licenses the total-variation bound of Proposition 1. Mixing the full moments does not satisfy this condition and therefore cannot carry the same guarantee, even when its accuracy is matched. The two constructions are tied in accuracy but not in the guarantees they provide; empirical equivalence in accuracy does not establish logical equivalence in the conditions each satisfies.

(2) On the value of the unification

[Tracked PDF: Pages 8–11, Lines 304–400]

We explicitly position the contribution as integration rather than a new operator. The unification is not a concatenation of existing techniques: it places feature-moment augmentation, convex-hull minimax, and weight averaging under one causal-invariance principle and states the conditions

under which each applies. This yields testable predictions—content–style separability as an enabling rather than sufficient condition, the behavior of the coverage gap, and the second-order flat-minima approximation. Together with the reproducible platform and the falsifiable boundary, this allows later methods to be compared on a common footing and avoids repeated investment in directions already shown to be ineffective. We do not claim a deployment advantage; the practical significance lies in the unified understanding and the well-characterized baseline.

(3) On the strength of the assumptions and the coverage term

[Tracked PDF: Page 8, Lines 304–322; Page 36, Lines 1171–1173]

The assumptions are now stated, tested, and falsifiable premises rather than hidden conditions. Content–style separability is labeled an enabling premise, and the trajectory–augmentation alignment is presented as an empirically supported assumption rather than a theorem, with its limitations stated. The target access required to evaluate the coverage-gap term is an inherent property of target-free generalization, not a limitation unique to this work; we demonstrate the behavior of both terms in a controlled moment-space setting (Figure 2, Page 14) rather than presenting the term as computable without target data. A strong assumption is not an incorrect one; the value of the analysis is that its conditions are explicit and open to falsification.

(4) On weakening claims versus adding evidence

[Tracked PDF: Pages 21–23, Lines 795–828; Pages 8–11, Lines 304–400]

When the evidence does not support a stronger claim, weakening that claim is the scientifically correct response; retaining an unsupported claim would be the error. The revision also adds positive evidence—an independent unified-code replication, a gate-pressure stress test, the alignment verification, and a corrected two-one-sided equivalence analysis. This evidence points to statistical equivalence and to boundary conditions rather than to superiority. What was weakened were the claims that did not hold; the claims that do hold—the unifying framework and the boundary characterization—are retained, strengthened, and made more precise and more readily falsifiable.

(5) Overall scientific significance

[Tracked PDF: Page 36, Lines 1168–1200]

The manuscript does not claim a performance breakthrough. Its scientific significance is at the level of understanding: a unifying explanation, a systematic validity boundary, and a falsifiable, reproducible benchmark; the transparent reporting of null and non-replicated results is itself part of that contribution. The only actionable editorial item has been completed (see the editorial cleanup below). We respectfully ask that the work be judged on what it actually establishes rather than on the overstated claims of the original submission.

(6) Editorial cleanup (duplicated or partially edited text)

[Tracked PDF: Page 37, Lines 1209–1211]

We carried out a full editorial review. The one concrete duplication was the data-availability statement, which appeared twice: a manually typeset "**Data availability.**" paragraph and the MDPI template `\dataavailability{}` command with identical content. The manual paragraph has been removed and the template command retained, so the statement now renders exactly once. The remaining items that can look like duplication in a text extraction (redefined revision macros, deliberate indentation and table alignment) were inspected individually and are not duplicated content.

Reviewer 3

General opening. We sincerely thank Reviewer 3 for the detailed and constructive assessment and for recognising the substantial improvement in the mathematical formulation, experimental validation, statistical analysis, and interpretation. We address the three limited revisions below.

R3-1. Mathematical justification for the conditional posterior bound

Comment. *Clarify the mathematical justification for transferring the total-variation bound to the conditional content posteriors. The data-processing inequality alone does not generally establish a pointwise posterior bound under different conditioning observations. A more explicit justification or an appropriately weakened statement would strengthen the theoretical validity.*

[Tracked PDF: Page 8, Line 304 through Page 9, Line 348; proofs on Page 10, Line 354 through Page 11, Line 400]

Response. The comment is correct, and we have restructured Proposition 1 rather than defend the earlier statement. The data-processing inequality controls the distributions (laws) of the posterior random variables, not the posterior evaluated at each fixed observation; at rare observations a small likelihood perturbation can be amplified. The proposition is now stated at two clearly separated levels:

- **(L2-i) Exact case.** When $\tau = 0$ and the partner source is exogenous, the content posterior is preserved exactly;
- **(L2-ii)(a) Distribution-level bounds (unconditional).** Without further assumptions, the coupling gives the only unconditional statements: the joint and observation-marginal laws

$$\mathrm{TV}(P_{C,Z}, P_{C,T(Z)}) \le \tau, \qquad \mathrm{TV}(p(Z), p(T(Z))) \le \tau;$$

- **(L2-ii)(b) Pointwise bound (conditional).** The pointwise statement is retained only under an explicit inter-model likelihood-ratio condition $|\log p(\mathbf{b}, r, \mathbf{d} \mid c) - \log p(\mathbf{b}, r, \mathbf{d}_{aug} \mid c)| \le \alpha$, under which $\mathrm{TV}(p(C \mid Z), p(C \mid T(Z))) \le \tanh \alpha$.

The KL corollary is now conditional on the same assumption, $D_{\mathrm{KL}} \le \tanh(\alpha)/q_{\min}$, and the earlier $\tau \log(M/p_{\min})$ and $\tau/q_{\min}$ pointwise bounds are removed. The label-posterior drift δ is treated as a directly measured quantity rather than added to τ. The full derivations, including the likelihood-ratio argument and the reverse-Pinsker inequality, are given in the revised proof (Pages 10–11). Theorems 1 and 2 are unaffected because they use the distribution-level total-variation quantity.

R3-2. Consistent distinction of equivalence, non-significance, and superiority

Comment. *The conclusions should consistently distinguish between demonstrated statistical equivalence, non-significant differences, and performance superiority. The conditional nature of the theoretical unification should also remain clear.*

[Tracked PDF: Page 1, Lines 17–19; Pages 21–23, Lines 795–828; Page 36, Lines 1168–1200]

Response. We have applied one consistent terminology throughout the twelve affected passages. Non-significant contrasts are reported as **statistically tied** or **on par** (no significant difference); the term **equivalence** is used only for the TOST procedure and always with the equivalence margin stated; and no superiority is claimed. We also corrected the TOST implementation to the standard two-one-sided procedure using 90% confidence intervals and recomputed every contrast. On PACS ($n = 10$), at the one-point margin equivalence is confirmed only against RSC; at the two-point margin it additionally holds against SWAD, CORAL, and Fish; the CIFA–EFDM contrast is neither superior nor certified equivalent. These wording changes appear at the abstract (Page 1), the introduction and contributions (Pages 2–3), the experiments and captions (Pages 21–25), the appendix (Pages 33–35), and the conclusions (Page 36). The conditional nature of the unification is kept explicit in every theoretical passage.

R3-3. Clean manuscript and consistency of equations, tables, references, and terminology

Comment. *Provide a clean manuscript incorporating the revisions and verify the consistency of equations, tables, cross-references, and terminology.*

[Tracked PDF: Whole manuscript; hyphenation at Page 36, Line 1168]

Response. A clean, fully accepted version (`CIFA_Electronics_clean.pdf`, 39 pages) is provided alongside the tracked version. We have verified that: every cross-reference resolves with no dangling or duplicate labels; all 27 bibliography entries are cited; equations, tables, and figures use automatic numbering with no hard-coded numbers; and the terminology and notation are consistent. The clean manuscript renders entirely in black with no revision marks. As part of this pass we also standardised the hyphenation (e.g., "content–style"). Both the clean and tracked manuscripts compile with zero errors and zero overfull boxes, and their accepted content is identical.

We hope these revisions fully address the reviewers' concerns and we thank the editorial team and reviewers once again for their time and guidance.

Sincerely,
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